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COLLEGE OF SCIENCE
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ADVENTURE WORKS PERFORMANCE ANALYTICS DASHBOARD
DATA IMPORT AND DATA MODELING DOCUMENTATION

By McReg Analytics

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ABOUT THE COMPANY

McReg Analytics is a data analytics company based in Ortigas, Pasig City, offering swift, efficient services to customers and striving to provide a bigger and brighter future. Our company guarantees professional, reliable services by transforming raw data into useful information that enables smarter decision-making. McReg Analytics assures customers that its team is highly competent, qualified, and dedicated to providing precise analytics. In addition, the company aims to assure customers that all information shared with McReg Analytics will be handled with the utmost discretion and confidentiality, processed securely, and used only for its intended purpose.

I. SOURCE DATA INVENTORY

Table 1 summarizes the profiling results for all 31 source tables selected from the original AdventureWorksDW2025 file before the implementation of exploratory data analysis and data modeling.

In implementing the statistical outlier analysis using the Interquartile range (IQR) method, numerous columns were flagged for variance. The team noticed that most of the outliers have explainable business meaning, but some weren't investigated further since it was outside of the project's scope:

1. YearlyIncome - 309 (1.67%) customers belong to extreme higher-income households.
2. NumberCarsOwned - 1, 261 (6.82%) customers belong to wealthier households, owning more than average.
3. DimEmployee - 14 (4.73%) employees, possibly executives/heads/specialists, earn above the median staff wage rate.
4. StandardCost, ListPrice, DealerPrice: 13 (3.29%)
5. Weight: 29 (10.28%) products (bike or huge bike accessories) weigh heavier than average.
6. DaysToManufacture: 125 (20.63%) products took longer to manufacture than average due to complexity.
7. MinQty: 2 (12.5%) orders have a higher order quantity minimum than average.
8. UnitPrice, ExtendedAmount, SalesAmount, TaxAmt, Freight: 9,586 (15.87%) internet sales orders shown higher-priced items (almost complete to complete bikes) compared to the average, with a majority of lower-mid priced bikes.
9. ProductStandardCost and TotalProductCost: 9,958 (16.49%) internet sales orders have a higher product cost than the average.



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10. UnitCost: 58336 (7.51%) of the products (bikes) in the inventory cost higher than average (accessories and small parts).
11. OrderQuantity: 5,192 (8.53%) reseller orders were bulk purchases with higher quantities than the typical single-unit retail order.
12. ExtendedAmount, SalesAmount, TaxAmt, Freight: 6,626–6,678 (10.89%–10.97%) reseller order lines carry higher revenue, tax, and shipping values, driven directly by the large-quantity orders in #11.
13. TotalProductCost: 6,338 (10.41%) reseller order lines have a higher cost basis, again tracking the same bulk orders.
14. ProductStandardCost: 3,411 (5.61%) reseller order lines involve products with a higher catalog standard cost, reflecting which products are being bought in bulk, not just how many.
15. UnitPrice: 1,852 (3.04%) reseller order lines show unusually high per-unit price — the smallest effect in this group, confirming quantity (not price) drives most reseller-side variance.
16. SalesAmountQuota: 1 (0.61%) quota record is unusually high or low compared to the rest.
17. YearlyIncome: 4 (0.19%) records show unusually high income of ProspectiveBuyer, meaning that only a small portion of prospective customers have higher reported income than the rest of the list.
18. NumberCarsOwned: 149 (7.24%) prospective buyers own more vehicles than average

As seen in Table 1, Row 23, FactFinance's Amount field shows a high outlier rate of 18.77%; however, Amount in this context combines multiple account types (adjustments to revenues and expenses), and therefore, the combined values significantly affect the variance. Since the project does not use FactFinance, as it is outside the scope of McReg Analytics Team's analysis and performance dashboard, this was only profiled, and no further investigations regarding the table structure of this attribute have occurred.

McReg Analytics therefore concludes that the dataset is referentially sound, with no duplicate keys or orphaned foreign key relationships identified, confirming that the dataset is pre-validated and a clean sample provided by Microsoft.

Table 1. Source Data Inventory

Table Name	Source Row Count	Data Types	Primary Key	Duplicates	Outliers <i>Format: attribute name, rows, percentage</i>	Referential Gaps
AdventureWorksDWBuildVersion	1	nvarchar, datetime	None declared	0	None flagged	None (no FKs declared)
DatabaseLog	96	int, datetime, nvarchar, xml	DatabaseLogID	0	None flagged	None (no FKs declared)
DimAccount	99	int, nvarchar	AccountKey	0	ParentAccountCodeAlternateKey: 4 (4.17%)	None found



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DimCurrency	105	int, nchar, nvarchar	CurrencyKey	0	None flagged	None (no FKs declared)
DimCustomer	18484	int, nvarchar, bit, date, nchar, money, tinyint	CustomerKey	0	YearlyIncome: 309 (1.67%); NumberCarsOwned: 1261 (6.82%)	None found
DimDate	3652	int, date, tinyint, nvarchar, smallint	DateKey	0	None flagged	None (no FKs declared)
DimDepartmentGroup	7	int, nvarchar	DepartmentGroupKey	0	None flagged	None found
DimEmployee	296	int, nvarchar, bit, date, nchar, tinyint, money, smallint, varbinary	EmployeeKey	0	BaseRate: 14 (4.73%)	None found
DimGeography	655	int, nvarchar	GeographyKey	0	None flagged	None found
DimOrganization	14	int, nvarchar	OrganizationKey	0	None flagged	None found
DimProduct	606	int, nvarchar, nchar, money, bit, smallint, float, varbinary, datetime	ProductKey	0	StandardCost: 13 (3.29%); ListPrice: 13 (3.29%); Weight: 29 (10.28%); DaysToManufacture: 125 (20.63%); DealerPrice: 13 (3.29%)	None found
DimProductCategory	4	int, nvarchar	ProductCategoryKey	0	None flagged	None (no FKs declared)
DimProductSubcategory	37	int, nvarchar	ProductSubcategoryKey	0	None flagged	None found
DimPromotion	16	int, nvarchar, float, datetime	PromotionKey	0	MinQty: 2 (12.5%)	None (no FKs declared)
DimReseller	701	int, nvarchar, varchar, char, tinyint, money	ResellerKey	0	None (OrderMonth got flagged but insignificant)	None found
DimSalesReason	10	int, nvarchar	SalesReasonKey	0	None flagged	None (no FKs declared)



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						declared)
DimSalesTerritory	11	int, nvarchar, varbinary	SalesTerritoryKey	0	None flagged	None (no FKs declared)
DimScenario	3	int, nvarchar	ScenarioKey	0	None flagged	None (no FKs declared)
FactAdditionalInter nationalProductDesc ription	1911	int, nvarchar	ProductKey + CultureName	0	None flagged	None (no FKs declared)
FactCallCenter	120	int, nvarchar, smallint, float, datetime	FactCallCenterID	0	None (TotalOperators, AverageTimePerIssue, and ServiceGrade were flagged but insignificant)	None found
FactCurrencyRate	14264	int, float, datetime	CurrencyKey + DateKey	0 duplicate key groups	None flagged	None found
FactFinance	39409	int, float, datetime	None declared	0 duplicate full-row groups (no PK declared)	Amount: 7399 (18.77%)	None found
FactInternetSales	60398	int, nvarchar, tinyint, smallint, money, float, datetime	SalesOrderNumber + SalesOrderLineNumber	0	UnitPrice: 9586 (15.87%); ExtendedAmount: 9586 (15.87%); ProductStandardCost: 9958 (16.49%); TotalProductCost: 9958 (16.49%); SalesAmount: 9586 (15.87%); TaxAmt: 9586 (15.87%); Freight: 9586 (15.87%)	None found
FactInternetSalesRe ason	64515	nvarchar, tinyint, int	SalesOrderNumber + SalesOrderLineNumber + SalesReasonKey	0	None flagged	None found



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FactProductInventory	776286	int, date, money	ProductKey + DateKey	0	UnitCost: 58336 (7.51%)	None found
FactResellerSales	60855	int, nvarchar, tinyint, smallint, money, float, datetime	SalesOrderNumber + SalesOrderLineNumber	0	OrderQuantity: 5192 (8.53%); UnitPrice: 1852 (3.04%); ExtendedAmount: 6678 (10.97%); ProductStandardCost: 3411 (5.61%); TotalProductCost: 6338 (10.41%); SalesAmount: 6626 (10.89%); TaxAmt: 6626 (10.89%); Freight: 6626 (10.89%)	None found
FactSalesQuota	163	int, smallint, tinyint, money, datetime	SalesQuotaKey	0	SalesAmountQuota: 1 (0.61%)	None found
FactSurveyResponse	2727	int, nvarchar, datetime	SurveyResponseKey	0	None flagged	None found
NewFactCurrencyRate	50	real, nvarchar, date, int	None declared	10 duplicate full-row groups (no PK declared)	None flagged	None (no FKs declared)
ProspectiveBuyer	2059	int, nvarchar, datetime, nchar, money, tinyint	ProspectiveBuyerKey	0	YearlyIncome: 4 (0.19%); NumberCarsOwned: 149 (7.24%)	None (no FKs declared)
sysdiagrams	9	nvarchar, int, varbinary	diagram id	0	None flagged	None (no FKs declared)

II. EXPLORATORY DATA ANALYSIS

The following steps are implemented in the dataset *AdventureWorksDW2025* file upon the execution of the exploratory data analysis.

Step 1: Load the data

```
RESTORE DATABASE [AdventureWorksDW2025]
FROM DISK = N'/var/opt/mssql/backup/AdventureWorksDW2025.bak'
```



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```
WITH MOVE N'AdventureWorksDW' TO N'/var/opt/mssql/data/AdventureWorksDW2025.mdf',  
MOVE N'AdventureWorksDW_log' TO N'/var/opt/mssql/data/AdventureWorksDW2025_log.ldf',  
STATS = 10;  
GO  
  
SELECT name FROM sys.databases;
```

Step 2: Overview of the Database

```
USE AdventureWorksDW2025;  
GO  
  
SELECT  
    t.name AS TableName,  
    p.rows AS [RowCount]  
FROM sys.tables t  
INNER JOIN sys.partitions p ON t.object_id = p.object_id AND p.index_id IN (0,1)  
ORDER BY t.name;
```

- This allows us to see the 31 tables that are within the database as well as the Row Count for each of the tables. This also serves as a sanity check whether we were able to successfully retrieve all of the data within the given database.

Step 3: Table Selection

The Full AdventureWorksDW2025 database contains 31 tables; however, not all tables are relevant in this project's scope. After much deliberation, several tables were concluded to just add noise and would not be able to contribute to the KPIs and business questions that the analysis would be built around. The following tables were excluded from the final model:

- AdventureWorksDWBuildVersion
- DatabaseLog
- Sysdiagrams
- DimAccount
- DimDepartmentGroup
- DimOrganization
- Dimscenario
- DimCurrency
- FactCurrencyRate
- NewFactCurrencyRate
- FactCallCenter



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- FactSurveyResponse
- FactFinance
- FactAdditionalInternationalProductDescription
- ProspectiveBuyer
- FactProductInventory

The following tables were retained for the model:

- DimCustomer
- DimDate
- DimEmployee
- DimGeography
- DimProduct
- DimProductSubCategory
- DimProductCategory
- DimPromotion
- DimReseller
- DimSalesReason
- DimSalesTerritory
- FactInternetSales
- FactInternetSalesReason
- FactResellerSales
- FactResellerQuota
- FactSalesQuota

Step 4: Merging Product, Subcategory and Category into a Single Table(DimProduct)

DimProduct → merges with DimProductSubCategory using ProductSubCategory Key → merges with DimProductCategory using ProductCategoryKey from the SubCategory table.

This is done in order to simplify the data model and reduce the number of relationships to make it easier to execute Power Queries in Power Bi.

```
SELECT COUNT(*) AS Orphaned_ProductSubcategoryKey
FROM DimProduct p
LEFT JOIN DimProductSubcategory s ON p.ProductSubcategoryKey = s.ProductSubcategoryKey
WHERE p.ProductSubcategoryKey IS NOT NULL AND s.ProductSubcategoryKey IS NULL;
```



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```
-- Confirm no unmatched category keys were lost in the merge
SELECT COUNT(*) AS Orphaned_ProductCategoryKey
FROM DimProductSubcategory s
LEFT JOIN DimProductCategory c ON s.ProductCategoryKey = c.ProductCategoryKey
WHERE c.ProductCategoryKey IS NULL;
```

- The result of this query was 0 which means that merge that happened was safe and every subcategory has a matching category.

Step 5: DimCustomer (Structure, Duplicates, and Nulls)

```
-- Structure: column names, data types, nullability
SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, CHARACTER_MAXIMUM_LENGTH
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'DimCustomer'
ORDER BY ORDINAL_POSITION;

-- Duplicate check on primary key
SELECT CustomerKey, COUNT(*) AS Occurrences
FROM DimCustomer
GROUP BY CustomerKey
HAVING COUNT(*) > 1;

SELECT
    SUM(CASE WHEN CustomerKey IS NULL THEN 1 ELSE 0 END) AS Null_CustomerKey,
    SUM(CASE WHEN GeographyKey IS NULL THEN 1 ELSE 0 END) AS Null_GeographyKey,
    SUM(CASE WHEN FirstName IS NULL THEN 1 ELSE 0 END) AS Null_FirstName,
    SUM(CASE WHEN LastName IS NULL THEN 1 ELSE 0 END) AS Null_LastName,
    SUM(CASE WHEN EmailAddress IS NULL THEN 1 ELSE 0 END) AS Null_EmailAddress,
    SUM(CASE WHEN BirthDate IS NULL THEN 1 ELSE 0 END) AS Null_BirthDate,
    SUM(CASE WHEN MaritalStatus IS NULL THEN 1 ELSE 0 END) AS Null_MaritalStatus,
    SUM(CASE WHEN Gender IS NULL THEN 1 ELSE 0 END) AS Null_Gender,
    SUM(CASE WHEN YearlyIncome IS NULL THEN 1 ELSE 0 END) AS Null_YearlyIncome,
    SUM(CASE WHEN TotalChildren IS NULL THEN 1 ELSE 0 END) AS Null_TotalChildren,
    SUM(CASE WHEN EnglishEducation IS NULL THEN 1 ELSE 0 END) AS Null_Education,
    SUM(CASE WHEN EnglishOccupation IS NULL THEN 1 ELSE 0 END) AS Null_Occupation,
    SUM(CASE WHEN HouseOwnerFlag IS NULL THEN 1 ELSE 0 END) AS Null_HouseOwnerFlag,
    SUM(CASE WHEN NumberCarsOwned IS NULL THEN 1 ELSE 0 END) AS Null_NumberCarsOwned,
    SUM(CASE WHEN DateFirstPurchase IS NULL THEN 1 ELSE 0 END) AS Null_DateFirstPurchase,
    COUNT(*) AS TotalRows
FROM DimCustomer;
```



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- After running this query, CustomerKey returned zero rows in the duplication check which confirms that each customer has a unique key and has been represented only once. For the null values check, it has returned zero null values. After this, we have carefully selected and renamed some of the columns in order to reduce data that are conveying the same information.

```
SELECT
    CustomerKey,
    GeographyKey,
    CONCAT(FirstName, ' ', LastName) AS FullName,
    EmailAddress,
    BirthDate,
    MaritalStatus,
    Gender,
    YearlyIncome,
    TotalChildren,
    NumberChildrenAtHome,
    EnglishEducation AS Education,
    EnglishOccupation AS Occupation,
    HouseOwnerFlag,
    NumberCarsOwned,
    DateFirstPurchase,
    CommuteDistance
FROM DimCustomer;
```

- We have decided to concatenate the first and last name into one single FullName column in order to uniquely identify each customer. Additionally, we only took the english versions of the occupations and educations as it is more universally used and people that will be analyzing this project are more familiar with the English language. For it to be even more universal, we have renamed it to just “Education” and “Occupation”.

Step 6: DimDate (Structure, Duplicates and Nulls)

```
-- Structure
SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, CHARACTER_MAXIMUM_LENGTH
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'DimDate'
ORDER BY ORDINAL_POSITION;

-- Duplicate check on primary key
SELECT DateKey, COUNT(*) AS Occurrences
FROM DimDate
GROUP BY DateKey
HAVING COUNT(*) > 1;
```



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```
-- Null value check
SELECT
  SUM(CASE WHEN DateKey IS NULL THEN 1 ELSE 0 END) AS Null_DateKey,
  SUM(CASE WHEN FullDateAlternateKey IS NULL THEN 1 ELSE 0 END) AS Null_FullDate,
  SUM(CASE WHEN DayNumberOfWeek IS NULL THEN 1 ELSE 0 END) AS Null_DayNumberOfWeek,
  SUM(CASE WHEN EnglishDayNameOfWeek IS NULL THEN 1 ELSE 0 END) AS Null_DayName,
  SUM(CASE WHEN MonthNumberOfYear IS NULL THEN 1 ELSE 0 END) AS Null_MonthNumber,
  SUM(CASE WHEN EnglishMonthName IS NULL THEN 1 ELSE 0 END) AS Null_MonthName,
  SUM(CASE WHEN CalendarQuarter IS NULL THEN 1 ELSE 0 END) AS Null_CalendarQuarter,
  SUM(CASE WHEN CalendarYear IS NULL THEN 1 ELSE 0 END) AS Null_CalendarYear,
  SUM(CASE WHEN FiscalQuarter IS NULL THEN 1 ELSE 0 END) AS Null_FiscalQuarter,
  SUM(CASE WHEN FiscalYear IS NULL THEN 1 ELSE 0 END) AS Null_FiscalYear,
  COUNT(*) AS TotalRows
FROM DimDate;
```

- There are zero duplicates on DateKey which confirms that there is one row per calendar day. Zero nulls as well across all date columns. Furthermore, in the code below, we have renamed some columns in order to simplify it even more.

```
SELECT
  DateKey,
  FullDateAlternateKey AS [Date],
  DayNumberOfWeek,
  EnglishDayNameOfWeek AS DayName,
  DayNumberOfMonth,
  MonthNumberOfYear,
  EnglishMonthName AS MonthName,
  CalendarQuarter,
  CalendarYear,
  CalendarSemester,
  FiscalQuarter,
  FiscalYear
FROM DimDate;
```

Step 7: DimEmployee (Structure, Duplicates and Nulls)

```
-- Structure
SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, CHARACTER_MAXIMUM_LENGTH
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'DimEmployee'
ORDER BY ORDINAL_POSITION;

-- Duplicate check on primary key
```



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```
SELECT EmployeeKey, COUNT(*) AS Occurrences
FROM DimEmployee
GROUP BY EmployeeKey
HAVING COUNT(*) > 1;

-- Null value check
SELECT
    SUM(CASE WHEN EmployeeKey IS NULL THEN 1 ELSE 0 END) AS Null_EmployeeKey,
    SUM(CASE WHEN FirstName IS NULL THEN 1 ELSE 0 END) AS Null_FirstName,
    SUM(CASE WHEN LastName IS NULL THEN 1 ELSE 0 END) AS Null_LastName,
    SUM(CASE WHEN Title IS NULL THEN 1 ELSE 0 END) AS Null_Title,
    SUM(CASE WHEN HireDate IS NULL THEN 1 ELSE 0 END) AS Null_HireDate,
    SUM(CASE WHEN SalesTerritoryKey IS NULL THEN 1 ELSE 0 END) AS Null_SalesTerritoryKey,
    SUM(CASE WHEN Gender IS NULL THEN 1 ELSE 0 END) AS Null_Gender,
    SUM(CASE WHEN DepartmentName IS NULL THEN 1 ELSE 0 END) AS Null_DepartmentName,
    COUNT(*) AS TotalRows
FROM DimEmployee;
```

- There are zero duplicates on EmployeeKey and zero nulls as well across all rows and columns. Furthermore, we have concatenated the first and last name of the employee names into one column named "FullName" to simplify it even more.

Step 8: DimGeography (Structure, Duplicates and Nulls)

```
-- Structure
SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, CHARACTER_MAXIMUM_LENGTH
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'DimGeography'
ORDER BY ORDINAL_POSITION;

-- Duplicate check on primary key
SELECT GeographyKey, COUNT(*) AS Occurrences
FROM DimGeography
GROUP BY GeographyKey
HAVING COUNT(*) > 1;

-- Null value check
SELECT
    SUM(CASE WHEN GeographyKey IS NULL THEN 1 ELSE 0 END) AS Null_GeographyKey,
    SUM(CASE WHEN City IS NULL THEN 1 ELSE 0 END) AS Null_City,
    SUM(CASE WHEN StateProvinceName IS NULL THEN 1 ELSE 0 END) AS Null_StateProvinceName,
    SUM(CASE WHEN EnglishCountryRegionName IS NULL THEN 1 ELSE 0 END) AS Null_Country,
```



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```
SUM(CASE WHEN PostalCode IS NULL THEN 1 ELSE 0 END) AS Null_PostalCode,  
SUM(CASE WHEN SalesTerritoryKey IS NULL THEN 1 ELSE 0 END) AS Null_SalesTerritoryKey,  
COUNT(*) AS TotalRows  
FROM DimGeography;
```

- There are zero duplicates on GeographyKey and zero nulls as well across all rows and columns. EnglishCountryRegionName has also been renamed to “Country”.

Step 9: DimProduct (Structure, Duplicates and Nulls)

```
-- Structure  
SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, CHARACTER_MAXIMUM_LENGTH  
FROM INFORMATION_SCHEMA.COLUMNS  
WHERE TABLE_NAME = 'DimProduct'  
ORDER BY ORDINAL_POSITION;  
  
-- Duplicate check on primary key  
SELECT ProductKey, COUNT(*) AS Occurrences  
FROM DimProduct  
GROUP BY ProductKey  
HAVING COUNT(*) > 1;  
  
-- Null value check  
SELECT  
    SUM(CASE WHEN ProductKey IS NULL THEN 1 ELSE 0 END) AS Null_ProductKey,  
    SUM(CASE WHEN EnglishProductName IS NULL THEN 1 ELSE 0 END) AS Null_ProductName,  
    SUM(CASE WHEN ProductSubcategoryKey IS NULL THEN 1 ELSE 0 END) AS Null_ProductSubcategoryKey,  
    SUM(CASE WHEN StandardCost IS NULL THEN 1 ELSE 0 END) AS Null_StandardCost,  
    SUM(CASE WHEN ListPrice IS NULL THEN 1 ELSE 0 END) AS Null_ListPrice,  
    SUM(CASE WHEN Color IS NULL THEN 1 ELSE 0 END) AS Null_Color,  
    SUM(CASE WHEN Size IS NULL THEN 1 ELSE 0 END) AS Null_Size,  
    SUM(CASE WHEN Weight IS NULL THEN 1 ELSE 0 END) AS Null_Weight,  
    SUM(CASE WHEN ProductLine IS NULL THEN 1 ELSE 0 END) AS Null_ProductLine,  
    SUM(CASE WHEN DealerPrice IS NULL THEN 1 ELSE 0 END) AS Null_DealerPrice,  
    SUM(CASE WHEN Class IS NULL THEN 1 ELSE 0 END) AS Null_Class,  
    SUM(CASE WHEN Style IS NULL THEN 1 ELSE 0 END) AS Null_Style,  
    SUM(CASE WHEN EndDate IS NULL THEN 1 ELSE 0 END) AS Null_EndDate_ActiveProducts,  
    COUNT(*) AS TotalRows  
FROM DimProduct;
```



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- There are zero duplicates on ProductKey. However, there were null values present in the columns of ListPrice, Color, Size and Weight which upon further investigation does not affect any result or queries that will be made in Power BI so we decided to just leave them be. In the EndDate column, there were null values wherein we have decided to just use the transform data option in PowerBI to fix this issue. There were also empty sections in ProductSubcategoryKey which also does not affect anything as these represent non-sellable items.

Step 10: DimPromotion (Structure, Duplicates and Nulls)

```
-- Structure
SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, CHARACTER_MAXIMUM_LENGTH
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'DimPromotion'
ORDER BY ORDINAL_POSITION;

-- Duplicate check on primary key
SELECT PromotionKey, COUNT(*) AS Occurrences
FROM DimPromotion
GROUP BY PromotionKey
HAVING COUNT(*) > 1;

-- Null value check
SELECT
    SUM(CASE WHEN PromotionKey IS NULL THEN 1 ELSE 0 END) AS Null_PromotionKey,
    SUM(CASE WHEN EnglishPromotionName IS NULL THEN 1 ELSE 0 END) AS Null_PromotionName,
    SUM(CASE WHEN DiscountPct IS NULL THEN 1 ELSE 0 END) AS Null_DiscountPct,
    SUM(CASE WHEN EnglishPromotionType IS NULL THEN 1 ELSE 0 END) AS Null_PromotionType,
    SUM(CASE WHEN EnglishPromotionCategory IS NULL THEN 1 ELSE 0 END) AS Null_PromotionCategory,
    SUM(CASE WHEN StartDate IS NULL THEN 1 ELSE 0 END) AS Null_StartDate,
    SUM(CASE WHEN EndDate IS NULL THEN 1 ELSE 0 END) AS Null_EndDate,
    COUNT(*) AS TotalRows
FROM DimPromotion;
```

- There are zero duplicates on PromotionKey and zero nulls as well across all rows and columns. The “No Discount” placeholder row will show a 0% discount rather than a null value so we did not do anything to change this. There were also some columns that have been renamed for simplicity.

Step 11: DimReseller (Structure, Duplicates and Nulls)

```
-- Structure
SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, CHARACTER_MAXIMUM_LENGTH
FROM INFORMATION_SCHEMA.COLUMNS
```



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```
WHERE TABLE_NAME = 'DimReseller'
ORDER BY ORDINAL_POSITION;

-- Duplicate check on primary key
SELECT ResellerKey, COUNT(*) AS Occurrences
FROM DimReseller
GROUP BY ResellerKey
HAVING COUNT(*) > 1;

-- Null value check
SELECT
    SUM(CASE WHEN ResellerKey IS NULL THEN 1 ELSE 0 END) AS Null_ResellerKey,
    SUM(CASE WHEN GeographyKey IS NULL THEN 1 ELSE 0 END) AS Null_GeographyKey,
    SUM(CASE WHEN ResellerName IS NULL THEN 1 ELSE 0 END) AS Null_ResellerName,
    SUM(CASE WHEN BusinessType IS NULL THEN 1 ELSE 0 END) AS Null_BusinessType,
    SUM(CASE WHEN AnnualSales IS NULL THEN 1 ELSE 0 END) AS Null_AnnualSales,
    SUM(CASE WHEN AnnualRevenue IS NULL THEN 1 ELSE 0 END) AS Null_AnnualRevenue,
    SUM(CASE WHEN NumberEmployees IS NULL THEN 1 ELSE 0 END) AS Null_NumberEmployees,
    SUM(CASE WHEN YearOpened IS NULL THEN 1 ELSE 0 END) AS Null_YearOpened,
    COUNT(*) AS TotalRows
FROM DimReseller;
```

- There are zero duplicates on ResellerKey and zero nulls as well across all rows and columns. There were also some columns that have been renamed for simplicity.

Step 12: DimSalesReason (Structure, Duplicates and Nulls)

```
-- Structure
SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, CHARACTER_MAXIMUM_LENGTH
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'DimSalesReason'
ORDER BY ORDINAL_POSITION;

-- Duplicate check on primary key
SELECT SalesReasonKey, COUNT(*) AS Occurrences
FROM DimSalesReason
GROUP BY SalesReasonKey
HAVING COUNT(*) > 1;

-- Null value check
SELECT
```



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```
SUM(CASE WHEN SalesReasonKey IS NULL THEN 1 ELSE 0 END) AS Null_SalesReasonKey,  
SUM(CASE WHEN SalesReasonName IS NULL THEN 1 ELSE 0 END) AS Null_SalesReasonName,  
SUM(CASE WHEN SalesReasonReasonType IS NULL THEN 1 ELSE 0 END) AS Null_SalesReasonType,  
COUNT(*) AS TotalRows  
FROM DimSalesReason;
```

- There are zero duplicates on SalesReasonKey and zero nulls as well across all rows and columns. There were also some columns that have been renamed for simplicity.

Step 13: DimSalesTerritory (Structure, Duplicates and Nulls)

```
-- Structure  
SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, CHARACTER_MAXIMUM_LENGTH  
FROM INFORMATION_SCHEMA.COLUMNS  
WHERE TABLE_NAME = 'DimSalesTerritory'  
ORDER BY ORDINAL_POSITION;  
  
-- Duplicate check on primary key  
SELECT SalesTerritoryKey, COUNT(*) AS Occurrences  
FROM DimSalesTerritory  
GROUP BY SalesTerritoryKey  
HAVING COUNT(*) > 1;  
  
-- Null value check  
SELECT  
    SUM(CASE WHEN SalesTerritoryKey IS NULL THEN 1 ELSE 0 END) AS Null_SalesTerritoryKey,  
    SUM(CASE WHEN SalesTerritoryRegion IS NULL THEN 1 ELSE 0 END) AS Null_Region,  
    SUM(CASE WHEN SalesTerritoryCountry IS NULL THEN 1 ELSE 0 END) AS Null_Country,  
    SUM(CASE WHEN SalesTerritoryGroup IS NULL THEN 1 ELSE 0 END) AS Null_Group,  
    COUNT(*) AS TotalRows  
FROM DimSalesTerritory;
```

- There are zero duplicates on SalesTerritoryKey and zero nulls as well across all rows and columns. There were also some columns that have been renamed for simplicity.

Step 14: FactInternetSales (Structure, Duplicates and Nulls)

```
-- Structure  
SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, CHARACTER_MAXIMUM_LENGTH  
FROM INFORMATION_SCHEMA.COLUMNS
```



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```
WHERE TABLE_NAME = 'FactInternetSales'
ORDER BY ORDINAL_POSITION;

-- Duplicate check on composite key
SELECT SalesOrderNumber, SalesOrderLineNumber, COUNT(*) AS Occurrences
FROM FactInternetSales
GROUP BY SalesOrderNumber, SalesOrderLineNumber
HAVING COUNT(*) > 1;

-- Null value check
SELECT
    SUM(CASE WHEN ProductKey IS NULL THEN 1 ELSE 0 END) AS Null_ProductKey,
    SUM(CASE WHEN CustomerKey IS NULL THEN 1 ELSE 0 END) AS Null_CustomerKey,
    SUM(CASE WHEN PromotionKey IS NULL THEN 1 ELSE 0 END) AS Null_PromotionKey,
    SUM(CASE WHEN SalesTerritoryKey IS NULL THEN 1 ELSE 0 END) AS Null_SalesTerritoryKey,
    SUM(CASE WHEN OrderDateKey IS NULL THEN 1 ELSE 0 END) AS Null_OrderDateKey,
    SUM(CASE WHEN DueDateKey IS NULL THEN 1 ELSE 0 END) AS Null_DueDateKey,
    SUM(CASE WHEN ShipDateKey IS NULL THEN 1 ELSE 0 END) AS Null_ShipDateKey,
    SUM(CASE WHEN OrderQuantity IS NULL THEN 1 ELSE 0 END) AS Null_OrderQuantity,
    SUM(CASE WHEN UnitPrice IS NULL THEN 1 ELSE 0 END) AS Null_UnitPrice,
    SUM(CASE WHEN SalesAmount IS NULL THEN 1 ELSE 0 END) AS Null_SalesAmount,
    SUM(CASE WHEN DiscountAmount IS NULL THEN 1 ELSE 0 END) AS Null_DiscountAmount,
    SUM(CASE WHEN TotalProductCost IS NULL THEN 1 ELSE 0 END) AS Null_TotalProductCost,
    SUM(CASE WHEN TaxAmt IS NULL THEN 1 ELSE 0 END) AS Null_TaxAmt,
    SUM(CASE WHEN Freight IS NULL THEN 1 ELSE 0 END) AS Null_Freight,
    COUNT(*) AS TotalRows
FROM FactInternetSales;

-- Referential integrity: orphaned foreign keys
SELECT COUNT(*) AS Orphaned_CustomerKey
FROM FactInternetSales f LEFT JOIN DimCustomer d ON f.CustomerKey = d.CustomerKey
WHERE d.CustomerKey IS NULL;

SELECT COUNT(*) AS Orphaned_ProductKey
FROM FactInternetSales f LEFT JOIN DimProduct d ON f.ProductKey = d.ProductKey
WHERE d.ProductKey IS NULL;

SELECT COUNT(*) AS Orphaned_PromotionKey
FROM FactInternetSales f LEFT JOIN DimPromotion d ON f.PromotionKey = d.PromotionKey
WHERE d.PromotionKey IS NULL;
```



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```
SELECT COUNT(*) AS Orphaned_OrderDateKey
FROM FactInternetSales f LEFT JOIN DimDate d ON f.OrderDateKey = d.DateKey
WHERE d.DateKey IS NULL;
```

- Zero duplicate order-line combinations confirms each row represents a unique sale line item. Zero nulls in the measure columns. Zero orphaned foreign keys across all four checks confirms every transaction correctly resolves to a real customer, product, promotion, and order date.

Step 15: FactInternetSalesReason (Structure, Duplicates and Nulls)

```
SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, CHARACTER_MAXIMUM_LENGTH
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'FactProductInventory'
ORDER BY ORDINAL_POSITION;

-- Preview the actual data
SELECT TOP 50 *
FROM FactProductInventory;

-- Row count
SELECT COUNT(*) AS TotalRows
FROM FactProductInventory;

-- Structure
SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, CHARACTER_MAXIMUM_LENGTH
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'FactInternetSalesReason'
ORDER BY ORDINAL_POSITION;

-- Duplicate check on composite key
SELECT SalesOrderNumber, SalesOrderLineNumber, SalesReasonKey, COUNT(*) AS Occurrences
FROM FactInternetSalesReason
GROUP BY SalesOrderNumber, SalesOrderLineNumber, SalesReasonKey
HAVING COUNT(*) > 1;

-- Null value check
SELECT
    SUM(CASE WHEN SalesOrderNumber IS NULL THEN 1 ELSE 0 END) AS Null_SalesOrderNumber,
    SUM(CASE WHEN SalesOrderLineNumber IS NULL THEN 1 ELSE 0 END) AS Null_SalesOrderLineNumber,
    SUM(CASE WHEN SalesReasonKey IS NULL THEN 1 ELSE 0 END) AS Null_SalesReasonKey,
```



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```
COUNT(*) AS TotalRows
FROM FactInternetSalesReason;

-- Referential integrity check
SELECT COUNT(*) AS Orphaned_SalesReasonKey
FROM FactInternetSalesReason f LEFT JOIN DimSalesReason d ON f.SalesReasonKey = d.SalesReasonKey
WHERE d.SalesReasonKey IS NULL;
```

- Zero duplicate order-line combinations confirms each row represents a unique sale line item. Zero nulls in the measure columns. Zero orphaned foreign keys across all checks confirms every transaction correctly resolves to a real customer.

Step 16: FactResellerSales (Structure, Duplicates and Nulls)

```
-- Structure
SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, CHARACTER_MAXIMUM_LENGTH
FROM INFORMATION_SCHEMA.COLUMNS
WHERE TABLE_NAME = 'FactResellerSales'
ORDER BY ORDINAL_POSITION;

-- Duplicate check on composite key
SELECT SalesOrderNumber, SalesOrderLineNumber, COUNT(*) AS Occurrences
FROM FactResellerSales
GROUP BY SalesOrderNumber, SalesOrderLineNumber
HAVING COUNT(*) > 1;

-- Null value check
SELECT
    SUM(CASE WHEN ProductKey IS NULL THEN 1 ELSE 0 END) AS Null_ProductKey,
    SUM(CASE WHEN ResellerKey IS NULL THEN 1 ELSE 0 END) AS Null_ResellerKey,
    SUM(CASE WHEN EmployeeKey IS NULL THEN 1 ELSE 0 END) AS Null_EmployeeKey,
    SUM(CASE WHEN SalesTerritoryKey IS NULL THEN 1 ELSE 0 END) AS Null_SalesTerritoryKey,
    SUM(CASE WHEN OrderDateKey IS NULL THEN 1 ELSE 0 END) AS Null_OrderDateKey,
    SUM(CASE WHEN OrderQuantity IS NULL THEN 1 ELSE 0 END) AS Null_OrderQuantity,
    SUM(CASE WHEN UnitPrice IS NULL THEN 1 ELSE 0 END) AS Null_UnitPrice,
    SUM(CASE WHEN SalesAmount IS NULL THEN 1 ELSE 0 END) AS Null_SalesAmount,
```



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```
SUM(CASE WHEN DiscountAmount IS NULL THEN 1 ELSE 0 END) AS Null_DiscountAmount,  
SUM(CASE WHEN TotalProductCost IS NULL THEN 1 ELSE 0 END) AS Null_TotalProductCost,  
COUNT(*) AS TotalRows  
FROM FactResellerSales;  
  
-- Referential integrity: orphaned foreign keys  
SELECT COUNT(*) AS Orphaned_ResellerKey  
FROM FactResellerSales f LEFT JOIN DimReseller d ON f.ResellerKey = d.ResellerKey  
WHERE d.ResellerKey IS NULL;  
  
SELECT COUNT(*) AS Orphaned_EmployeeKey  
FROM FactResellerSales f LEFT JOIN DimEmployee d ON f.EmployeeKey = d.EmployeeKey  
WHERE d.EmployeeKey IS NULL;  
  
SELECT COUNT(*) AS Orphaned_ProductKey  
FROM FactResellerSales f LEFT JOIN DimProduct d ON f.ProductKey = d.ProductKey  
WHERE d.ProductKey IS NULL;
```

- Zero duplicate order-line combinations confirms each row represents a unique sale line item. Zero nulls in the measure columns. Zero orphaned foreign keys across all checks confirms every transaction correctly resolves to a real customer.

Step 17: FactSalesQuota (Structure, Duplicates and Nulls)

```
-- Structure  
SELECT COLUMN_NAME, DATA_TYPE, IS_NULLABLE, CHARACTER_MAXIMUM_LENGTH  
FROM INFORMATION_SCHEMA.COLUMNS  
WHERE TABLE_NAME = 'FactSalesQuota'  
ORDER BY ORDINAL_POSITION;  
  
-- Duplicate check on composite key  
SELECT EmployeeKey, CalendarYear, CalendarQuarter, COUNT(*) AS Occurrences  
FROM FactSalesQuota  
GROUP BY EmployeeKey, CalendarYear, CalendarQuarter  
HAVING COUNT(*) > 1;  
  
-- Null value check
```



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```
SELECT
    SUM(CASE WHEN EmployeeKey IS NULL THEN 1 ELSE 0 END) AS Null_EmployeeKey,
    SUM(CASE WHEN CalendarYear IS NULL THEN 1 ELSE 0 END) AS Null_CalendarYear,
    SUM(CASE WHEN CalendarQuarter IS NULL THEN 1 ELSE 0 END) AS Null_CalendarQuarter,
    SUM(CASE WHEN SalesAmountQuota IS NULL THEN 1 ELSE 0 END) AS Null_SalesAmountQuota,
    COUNT(*) AS TotalRows
FROM FactSalesQuota;

-- Referential integrity check
SELECT COUNT(*) AS Orphaned_EmployeeKey
FROM FactSalesQuota f LEFT JOIN DimEmployee d ON f.EmployeeKey = d.EmployeeKey
WHERE d.EmployeeKey IS NULL;
```

- Zero duplicate order-line combinations confirms each row represents a unique sale line item. Zero nulls in the measure columns. Zero orphaned foreign keys across all checks confirms every transaction correctly resolves to a real customer.

This table serves as the final EDA checkpoint before moving into data modeling in Power BI: it confirms every retained table is populated, free of duplicate keys, free of unexpected nulls in critical fields, and fully referentially intact against its related dimensions

III. DATA DICTIONARY

Source System: AdventureWorksDW2025

Shown in Table 2 is the Data Dictionary of the 13 tables (originally 31 tables), which identifies the table names, their respective attributes, business definitions, types, formats, range, if it's a required field, PK or FK, and its FK-referenced table.

Table 2. Data Dictionary

TABLE NAME	ATTRIBUTE NAME	BUSINESS DEFINITION	TYPE	FORMAT	RANGE	REQUIRED	PK or FK	FK REFERENCED TABLE
DimCustomer	CustomerKey	Unique primary identifier for a customer record.	INT	Integer	11000 to 29483	Yes	PK	N/A



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	GeographyKey	Foreign key referencing the geographic location details.	INT	Integer	2 to 654	No	FK	DimGeography
	FullName	Given name of the individual.	NVARCHAR	Text	Aaron to Zoe Adams to Zukowski	No	No	N/A
	MaritalStatus	Marital status of the individual (e.g., Single, Married).	NCHAR	Text	M to S	No	No	N/A
	Gender	Gender identity of the individual (e.g., M, F).	NVARCHAR	Text	F to M	No	No	N/A
	EmailAddress	Primary contact email address.	NVARCHAR	Text	aaron10@adventure-works.com to zoe9@adventure-works.com	No	No	N/A
	YearlyIncome	Estimated annual household income of the customer.	MONEY	Currency / Decimal	10000.0000 to 170000.0000	No	No	N/A
	TotalChildren	Total count of children associated with the customer.	TINYINT	Integer	0 to 5	No	No	N/A
	NumberChildrenAtHome	Number of children currently living in the customer household.	TINYINT	Integer	0 to 5	No	No	N/A
	Education	Educational attainment level described in English.	NVARCHAR	Text	Bachelors to Partial High School	No	No	N/A



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	Occupation	Primary occupation category described in English.	NVAR CHAR	Text	Clerical to Skilled Manual	No	No	N/A
	HouseOwnerFlag	Indicator whether the customer owns a home (1) or rents (0).	NCH AR	Text	0 to 1	No	No	N/A
	NumberCarsOwned	Total number of motor vehicles owned by the customer.	TINYI NT	Integer	0 to 4	No	No	N/A
	DateFirstPurchase	Date when the customer made their initial purchase.	DATE	YYYY-MM-DD	2010-12-29 to 2014-01-28	No	No	N/A
	CommuteDistance	Estimated daily travel distance to work.	NVAR CHAR	Text	0-1 Miles to 5-10 Miles	No	No	N/A
DimDate	DateKey	Surrogate key representing a specific date in YYYYMMDD format.	INT	Integer	20050101 to 20141231	Yes	PK	N/A
	DayNumberOfWeek	Numeric index representing the day of the week.	TINYI NT	Integer	1 to 7	Yes	No	N/A
	DayName	Name of the day of the week in English.	NVAR CHAR	Text	Friday to Wednesday	Yes	No	N/A
	DayNumberOfMonth	Day number within the calendar month.	TINYI NT	Integer	1 to 31	Yes	No	N/A
	DayNumberOfYear	Sequential day count within the calendar year.	SMAL LINT	Integer	1 to 366	Yes	No	N/A



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	MonthName	Name of the calendar month in English.	NVAR CHAR	Text	April to September	Yes	No	N/A
	MonthNumberOfYear	Numeric month of the year (1 to 12).	TINYI NT	Integer	1 to 12	Yes	No	N/A
	CalendarQuarter	Calendar quarter index (1 to 4).	TINYI NT	Integer	1 to 4	Yes	No	N/A
	CalendarYear	Four-digit calendar year.	SMAL LINT	Integer	2005 to 2014	Yes	No	N/A
	CalendarSemester	Calendar half-year period (1 or 2).	TINYI NT	Integer	1 to 2	Yes	No	N/A
	FiscalQuarter	Organization fiscal quarter index (1 to 4).	TINYI NT	Integer	1 to 4	Yes	No	N/A
	FiscalYear	Organization fiscal accounting year.	SMAL LINT	Integer	2005 to 2014	Yes	No	N/A
DimEmployee	EmployeeKey	Unique primary identifier for an employee.	INT	Integer	1 to 296	Yes	PK	N/A
	SalesTerritoryKey	Foreign key referencing the assigned sales territory.	INT	Integer	1 to 11	No	FK	hSalesTerrit
	FullName	Given name of the individual.	NVAR CHAR	Text	A. Scott to Zheng Abbas to Zwilling	Yes	No	N/A
	Title	Honorific title or prefix for the individual.	NVAR CHAR	Text	Accountant to Vice President of Sales	No	No	N/A
	HireDate	Official date of employment commencement.	DATE	YYYY-MM-DD	2006-01-28 to 2012-12-28	No	No	N/A



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	MaritalStatus	Marital status of the individual (e.g., Single, Married).	NCHAR	Text	M to S	No	No	N/A
	SalariedFlag	Indicator whether employee is paid salary (1) or hourly (0).	BIT	Boolean (0/1)	N/A	No	No	N/A
	Gender	Gender identity of the individual (e.g., M, F).	NCHAR	Text	F to M	No	No	N/A
	CurrentFlag	Indicator whether the employee is currently active (1) or inactive (0).	BIT	Boolean (0/1)	N/A	Yes	No	N/A
	SalesPersonFlag	Indicator whether the employee is designated as a sales staff member.	BIT	Boolean (0/1)	N/A	Yes	No	N/A
	DepartmentName	Organizational business unit or department name.	NVARCHAR	Text	Document Control to Tool Design	No	No	N/A
	StartDate	Effective start date of the record or assignment.	DATE	YYYY-MM-DD	2006-01-28 to 2013-06-13	No	No	N/A
	EndDate	Effective end date of the record or assignment.	DATE	YYYY-MM-DD	2009-02-11 to 2013-06-13	No	No	N/A
	Status	Current operational employment status.	NVARCHAR	Text	Current	No	No	N/A
DimGeography	GeographyKey	Foreign key referencing the	INT	Integer	1 to 655	Yes	PK	N/A



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		geographic location details.						
	City	Name of the municipality or city.	NVAR CHAR	Text	Abingdon to Zeeland	No	No	N/A
	StateProvinceName	Full official name of the state or province.	NVAR CHAR	Text	Alabama to Yveline	No	No	N/A
	CountryRegionCode	ISO standard country or region code.	NVAR CHAR	Text	AU to US	No	No	N/A
	Country	Name of the country or region in English.	NVAR CHAR	Text	Australia to United States	No	No	N/A
	PostalCode	Postal or ZIP code of the geographic region.	NVAR CHAR	Text	01071 to YO15	No	No	N/A
	SalesTerritoryKey	Foreign key referencing the assigned sales territory.	INT	Integer	1 to 10	No	FK	DimSalesTerritory
DimProduct	ProductKey	Column containing ProductKey	INT	Integer	1 to 606	Yes	PK	N/A
	ProductSKU	Column containing ProductAlternateKey	NVAR CHAR	Text	AR-5381 to WB-H098	No	No	N/A
	ProductName	Column containing EnglishProductName	NVAR CHAR	Text	Adjustable Race to Women's Tights, S	Yes	No	N/A
	StandardCost	Column containing StandardCost	MON EY	Currency / Decimal	0.8565 to 2171.2942	No	No	N/A
	Color	Column containing Color	NVAR CHAR	Text	Black to Yellow	Yes	No	N/A
	ReorderPoint	Column containing ReorderPoint	SMAL LINT	Integer	3 to 750	No	No	N/A



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	ListPrice	Column containing ListPrice	MON EY	Currency / Decimal	2.2900 to 3578.2700	No	No	N/A
	Size	Column containing Size	NVAR CHAR	Text	38 to XL	No	No	N/A
	Weight	Column containing Weight	FLOA T	Currency / Decimal	2.12 to 1050.0	No	No	N/A
	ProductLine	Column containing ProductLine	NCH AR	Text	M to T	No	No	N/A
	DealerPrice	Column containing DealerPrice	MON EY	Currency / Decimal	1.3740 to 2146.9620	No	No	N/A
	Class	Column containing Class	NCH AR	Text	H to M	No	No	N/A
	Style	Column containing Style	NCH AR	Text	M to W	No	No	N/A
	ModelName	Column containing ModelName	NVAR CHAR	Text	All-Purpose Bike Stand to Women's Tights	No	No	N/A
	Description	Column containing EnglishDescription	NVAR CHAR	Text	A light yet stiff aluminum bar for long distance riding. to Wraps to fit front and rear tires, carrier and 2 keys included.	No	No	N/A
	Status	Column containing Status	NVAR CHAR	Text	Current	No	No	N/A
	ProductSubcategoryKey	Column containing ProductSubcategoryKey	INT	Integer	1 to 37	No	No	N/A



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	ProductSubcategoryName	Column containing EnglishProductSubcategoryName	NVAR CHAR	Text	Bib-Shorts to Wheels	No	No	N/A
	ProductCategoryKey	Column containing ProductCategoryKey	INT	Integer	1 to 4	No	No	N/A
	ProductCategoryName	Column containing EnglishProductCategoryName	NVAR CHAR	Text	Accessories to Components	No	No	N/A
DimPromotion	PromotionKey	Unique surrogate identifier for a promotional offer.	INT	Integer	1 to 16	Yes	PK	N/A
	PromotionName	Descriptive name of the promotion in English.	NVAR CHAR	Text	Half-Price Pedal Sale to Volume Discount over 60	No	No	N/A
	DiscountPct	Percentage discount rate applied during promotion.	FLOAT	Currency / Decimal	0.0 to 0.5	No	No	N/A
	PromotionType	Classification category of promotion in English.	NVAR CHAR	Text	Discontinued Product to Volume Discount	No	No	N/A
	PromotionCategory	Target audience category of promotion in English.	NVAR CHAR	Text	Customer to Reseller	No	No	N/A
	StartDate	Effective start date of the record or assignment.	DATE TIME	YYYY-MM-DD	2010-11-29 00:00:00 to 2013-10-29 00:00:00	Yes	No	N/A
	EndDate	Effective end date of the record or assignment.	DATE TIME	YYYY-MM-DD	2011-12-28 00:00:00 to 2014-06-30 00:00:00	No	No	N/A



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DimReseller	ResellerKey	Unique primary identifier for a reseller business entity.	INT	Integer	1 to 701	Yes	PK	N/A
	GeographyKey	Foreign key referencing the geographic location details.	INT	Integer	1 to 655	No	FK	DimGeography
	BusinessType	Classification type of reseller business entity (e.g., Value Added Reseller, Warehouse).	VARC HAR	Text	Specialty Bike Shop to Warehouse	Yes	No	N/A
	ResellerName	Official commercial business name of the reseller.	NVAR CHAR	Text	A Bicycle Association to Yellow Bicycle Company	Yes	No	N/A
	NumberEmployees	Total headcount of employees employed by the reseller.	INT	Integer	2 to 100	No	No	N/A
	ProductLine	Primary product category line serviced by the reseller.	NVAR CHAR	Text	Mountain to Touring	No	No	N/A
	AnnualSales	Total historical annual sales volume processed through reseller.	MON EY	Currency / Decimal	300000.0000 to 3000000.0000	No	No	N/A
	AnnualRevenue	Total annual monetary revenue reported by reseller entity.	MON EY	Currency / Decimal	30000.0000 to 300000.0000	No	No	N/A
	YearOpened	Calendar year in which the reseller business	INT	Integer	1974 to 2005	No	No	N/A



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		was founded or established.						
DimSalesReason	SalesReasonKey	Unique identifier for a business sales reason.	INT	Integer	1 to 10	Yes	PK	N/A
	SalesReasonName	Descriptive label of the customer purchase reason.	NVAR CHAR	Text	Demo Event to Television Advertisement	Yes	No	N/A
	SalesReasonReasonType	High-level categorization category of the sales reason.	NVAR CHAR	Text	Marketing to Promotion	Yes	No	N/A
DimSalesTerritory	SalesTerritoryKey	Foreign key referencing the assigned sales territory.	INT	Integer	1 to 11	Yes	PK	N/A
	SalesTerritoryRegion	Geographic regional boundary name of the territory.	NVAR CHAR	Text	Australia to United Kingdom	Yes	No	N/A
	SalesTerritoryCountry	Country boundary name for the assigned sales territory.	NVAR CHAR	Text	Australia to United States	Yes	No	N/A
	SalesTerritoryGroup	Macro-geographic territory grouping (e.g., North America, Europe).	NVAR CHAR	Text	Europe to Pacific	No	No	N/A
FactInternetSales	ProductKey	Foreign key referencing the product dimensional record.	INT	Integer	214 to 606	Yes	FK	DimProduct
	OrderDateKey	Foreign key referencing the date the order was placed.	INT	Integer	20101229 to 20140128	Yes	FK	DimDate



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	DueDateKey	Foreign key referencing the payment or fulfillment due date.	INT	Integer	20110110 to 20140209	Yes	FK	DimDate
	ShipDateKey	Foreign key referencing the date product shipment occurred.	INT	Integer	20110105 to 20140204	Yes	FK	DimDate
	CustomerKey	Unique primary identifier for a customer record.	INT	Integer	11000 to 29483	Yes	FK	DimCustomer
	PromotionKey	Unique surrogate identifier for a promotional offer.	INT	Integer	1 to 14	Yes	FK	DimPromotion
	SalesTerritoryKey	Foreign key referencing the assigned sales territory.	INT	Integer	1 to 10	Yes	FK	DimSalesTerritory
	SalesOrderNumber	Unique system document identification number for the sales order.	NVARCHAR	Text	SO43697 to SO75123	Yes	PK	N/A
	SalesOrderLineNumber	Line item sequence number within a specific sales order.	TINYINT	Integer	1 to 8	Yes	PK	N/A
	OrderQuantity	Total unit quantity of products ordered in transaction line.	SMALLINT	Integer	1	Yes	No	N/A



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	UnitPrice	Selling price per individual unit charged on order line.	MON EY	Currency / Decimal	2.2900 to 3578.2700	Yes	No	N/A
	ExtendedAmount	Total order line value before discounts (OrderQuantity multiplied by UnitPrice).	MON EY	Currency / Decimal	2.2900 to 3578.2700	Yes	No	N/A
	UnitPriceDiscountPct	Percentage discount applied to individual unit price.	FLOA T	Currency / Decimal	0	Yes	No	N/A
	DiscountAmount	Total monetary discount value applied on transaction line.	FLOA T	Currency / Decimal	0	Yes	No	N/A
	ProductStandardCost	Standard production or manufacturing acquisition cost per unit.	MON EY	Currency / Decimal	0.8565 to 2171.2942	Yes	No	N/A
	TotalProductCost	Total standard cost for line item (OrderQuantity multiplied by ProductStandardCost).	MON EY	Currency / Decimal	0.8565 to 2171.2942	Yes	No	N/A
	SalesAmount	Net sales monetary revenue credited for line item after discounts.	MON EY	Currency / Decimal	2.2900 to 3578.2700	Yes	No	N/A
	TaxAmt	Total government sales tax charge levied on order line.	MON EY	Currency / Decimal	0.1832 to 286.2616	Yes	No	N/A



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	Freight	Shipping, logistics, and delivery charge applied to transaction line.	MON EY	Currency / Decimal	0.0573 to 89.4568	Yes	No	N/A
FactInternetSalesReason	SalesOrderNumber	Unique system document identification number for the sales order.	NVAR CHAR	Text	SO43697 to SO75123	Yes	PK	FactInternetSalesReason
	SalesOrderLineNumber	Line item sequence number within a specific sales order.	TINYI NT	Integer	1 to 8	Yes	PK	FactInternetSalesReason
	SalesReasonKey	Unique identifier for a business sales reason.	INT	Integer	1 to 10	Yes	PK	FactInternetSalesReason
FactResellerSales	ProductKey	Foreign key referencing the product dimensional record.	INT	Integer	212 to 606	Yes	FK	DimProduct
	OrderDateKey	Foreign key referencing the date the order was placed.	INT	Integer	20101229 to 20131129	Yes	FK	DimDate
	DueDateKey	Foreign key referencing the payment or fulfillment due date.	INT	Integer	20110110 to 20131211	Yes	FK	DimDate
	ShipDateKey	Foreign key referencing the date product shipment occurred.	INT	Integer	20110105 to 20131206	Yes	FK	DimDate
	ResellerKey	Unique primary identifier for a reseller business entity.	INT	Integer	1 to 701	Yes	FK	DimReseller



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	EmployeeKey	Unique primary identifier for an employee.	INT	Integer	272 to 296	Yes	FK	DimEmployee
	PromotionKey	Unique surrogate identifier for a promotional offer.	INT	Integer	1 to 16	Yes	FK	DimPromotion
	SalesTerritoryKey	Foreign key referencing the assigned sales territory.	INT	Integer	1 to 10	Yes	FK	DimSalesTerritory
	SalesOrderNumber	Unique system document identification number for the sales order.	NVARCHAR	Text	SO43659 to SO71952	Yes	PK	N/A
	SalesOrderLineNumber	Line item sequence number within a specific sales order.	TINYINT	Integer	1 to 72	Yes	PK	N/A
	OrderQuantity	Total unit quantity of products ordered in transaction line.	SMALLINT	Integer	1 to 44	No	No	N/A
	UnitPrice	Selling price per individual unit charged on order line.	MONEY	Currency / Decimal	1.3282 to 2146.9620	No	No	N/A
	ExtendedAmount	Total order line value before discounts (OrderQuantity multiplied by UnitPrice).	MONEY	Currency / Decimal	1.3740 to 30992.9100	No	No	N/A



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	UnitPriceDiscountPct	Percentage discount applied to individual unit price.	FLOAT	Currency / Decimal	0.0 to 0.4	No	No	N/A
	DiscountAmount	Total monetary discount value applied on transaction line.	FLOAT	Currency / Decimal	0.0 to 4005.2376	No	No	N/A
	ProductStandardCost	Standard production or manufacturing acquisition cost per unit.	MONEY	Currency / Decimal	0.8565 to 2171.2942	No	No	N/A
	TotalProductCost	Total standard cost for line item (OrderQuantity multiplied by ProductStandardCost).	MONEY	Currency / Decimal	0.8565 to 38530.3854	No	No	N/A
	SalesAmount	Net sales monetary revenue credited for line item after discounts.	MONEY	Currency / Decimal	1.3740 to 27893.6190	No	No	N/A
	TaxAmt	Total government sales tax charge levied on order line.	MONEY	Currency / Decimal	0.1099 to 2231.4895	No	No	N/A
	Freight	Shipping, logistics, and delivery charge applied to transaction line.	MONEY	Currency / Decimal	0.0344 to 697.3405	No	No	N/A
FactSalesQuota	SalesQuotaKey	Unique primary key for a sales quota baseline entry.	INT	Integer	1 to 163	Yes	PK	N/A

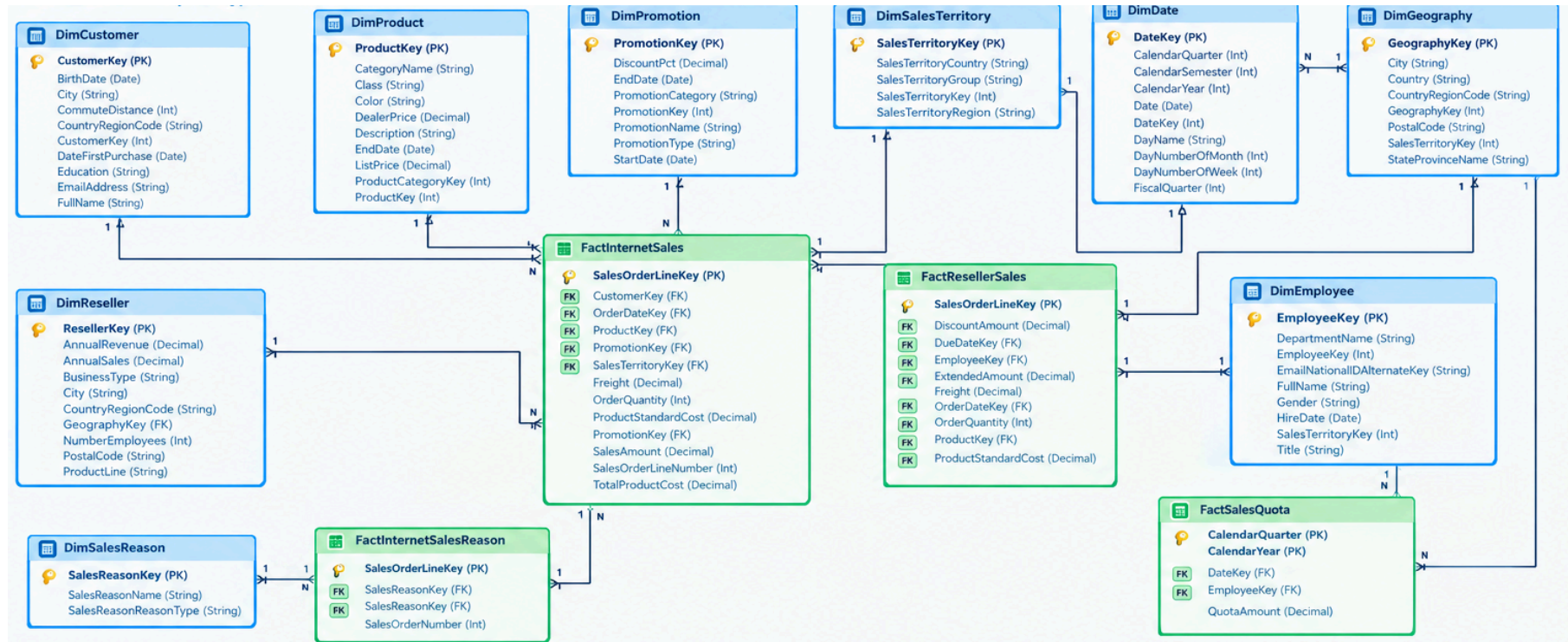


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	EmployeeKey	Unique primary identifier for an employee.	INT	Integer	272 to 296	Yes	FK	DimEmployee
	DateKey	Surrogate key representing a specific date in YYYYMMDD format.	INT	Integer	20101229 to 20130929	Yes	FK	DimDate
	CalendarYear	Four-digit calendar year.	SMALLINT	Integer	2010 to 2013	Yes	No	N/A
	CalendarQuarter	Calendar quarter index (1 to 4).	TINYINT	Integer	1 to 4	Yes	No	N/A
	SalesAmountQuota	Targeted monetary sales revenue quota benchmark for period.	MONEY	Currency / Decimal	1000.0000 to 1898000.0000	Yes	No	N/A

IV. ENTITY RELATIONSHIP DIAGRAM (ERD)



- We have used a Constellation Schema which is a group of Star Schemas as our model in this project because it makes it easier in Power BI to do DAX queries because it goes through less tables in order to compute. The following are the business rules:
 - Customer and Internet Sales
 - A customer can have many internet sales transactions.
 - Each internet sales transaction belongs to one customer.
 - A customer is identified uniquely by “CustomerKey”.
 - Date and Sales
 - A date can be associated with many internet sales transactions.
 - Each internet sales transaction is associated with one date through “OrderDateKey”.



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- A date can be associated with many reseller sales transactions.
- Each reseller sales transaction is associated with one date through “OrderDateKey”.
- “DimDate” provides the calendar information used to analyze sales by day, month, quarter, semester, and year.
- c. Product and Sales
 - A product can appear in many internet sales transactions.
 - Each internet sales transaction refers to one product.
 - A product can appear in many reseller sales transactions.
 - Each reseller sales transaction refers to one product.
 - A product is uniquely identified by “ProductKey”
- d. Promotion and Internet Sales
 - A promotion can be associated with many internet sales transactions.
 - Each internet sales transaction can be associated with one promotion through “PromotionKey”
 - Promotion information includes the promotion category, type, discount, and validity dates.
- e. Sales Territory
 - A sales territory can have many internet sales transactions.
 - Each internet sales transaction belongs to one sales territory.
 - A sales territory can have many reseller sales transactions.
 - Each reseller sales transaction belongs to one sales territory.
 - A sales territory is identified by “SalesTerritoryKey”.
- f. Reseller and Reseller Sales
 - A reseller can have many reseller sales transactions.
 - Each reseller sales transaction is associated with one reseller.
 - A reseller is uniquely identified by “ResellerKey”.
 - Reseller information includes business type, location, revenue, sales, and number of employees.
- g. Employee and Reseller Sales
 - An employee can be associated with many reseller sales transactions.
 - Each reseller sales transaction is associated with one employee through “EmployeeKey”.
 - An employee is uniquely identified by “EmployeeKey”.
- h. Sales Reasons
 - A sales reason can be associated with many internet sales reason records.
 - Each internet sales reason record refers to one sales reason.
 - An internet sales transaction can have multiple associated sales reasons.
 - A sales reason is uniquely identified by “SalesReasonKey”.



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i. Sales Quota

- An employee can have many sales quota records.
- Each sales quota record belongs to one employee.
- A date can have many sales quota records.
- Each sales quota record is associated with one date.
- Sales quota information can therefore be analyzed by employee and date.

V. DECLARATION ON USAGE OF AI

Generative AI tools (specifically Claude and Gemini) were used to assist in coding and filling out the Business Definition of some attributes for the AdventureWorksDW2025 Data Dictionary due to the large number of columns (attributes) within the entire dataset. However, McReg Analytics guarantees that the final content was reviewed, verified, and edited independently by our team, and we take full responsibility for what is placed within the Data Import and Data Modeling Documentation.